

Exam Date & Time: 31-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech VII Semester - Make up Examination - December -2024
software Engineering [CSE-4042]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Construct a schematic diagram to represent the iterative waterfall model of software development. On the diagram represents the phase entry and exit criteria for each phase. (5)
 - A)
 - B) A startup wants to launch an e-commerce website. However, they are uncertain about the exact features customers will need and anticipate many changes based on feedback. Apply the agile model with an iterative waterfall. (3)
 - C) A bank wants to build a new online banking portal for customers to check balances, transfer funds, and view transaction history. Identify the importance of the Design phase in the SDLC. (2)
- 2) You are assigned to gather requirements for a hospital management system. Apply the requirements-gathering techniques and analyze how you would ensure all stakeholders' needs are identified and documented accurately. (5)
 - A)
 - B) You are tasked with preparing a Software Requirements Specification (SRS) document for a library management system. Analyze the necessary components of an SRS document to ensure clarity and completeness. Briefly describe how you would organize the document. (3)
 - C) You are tasked with designing a website for an e-commerce platform. Apply two key steps of the design process to ensure the website is both functional and visually appealing. (2)
- 3) A team has developed two different software designs for a financial management system. One design uses a monolithic architecture, and the other uses a microservices architecture. Analyze the advantages and disadvantages of each design approach in terms of scalability, maintainability, and performance. Which design would you recommend for a large organization and why? (5)
 - A)
 - B) Explain the concept of a layered arrangement of modules in software design. How does this approach enhance system maintainability and flexibility? (3)
 - C) (2)

Analyze the role of use case diagrams in clarifying software requirements. How do they contribute to a better understanding of user interactions?

- 4) In a hospital's patient management system, patients can schedule appointments, doctors can access patient records, and billing information is processed. Draw a DFD representing the patient appointment scheduling process. What information flows between the patients, doctors, and billing department? (5)
- A) **Considerations:**
- Identify processes such as "Schedule Appointment," "Access Patient Records," and "Process Billing."
 - Determine external entities like "Patient," "Doctor," and "Billing Department."
- Include relevant data stores like "Patient Records" and "Appointment Schedule."
- B) Construct the use-case diagram "Withdraw the Cash from ATM" by identifying appropriate synchronous and asynchronous messages, object creation deletion and fragments. (3)
- C) A restaurant wants to build an online food ordering platform where customers can place orders, and view their status, and restaurants can manage the menu and delivery process. Identify how can Object-Oriented Analysis and Design (OOAD) be applied in Online Food Ordering System development. (2)
- 5) You are conducting performance testing for a new mobile application. Apply the steps of performance testing to evaluate the app's ability to handle high traffic and large data volumes. What would you test, and how would you ensure the app meets performance expectations? (5)
- A)
- B) You are working on a large software project and want to identify potential performance bottlenecks. Apply a program analysis tool to explain how you would use it to detect inefficiencies in the code. (3)
- C) In a web application, logging errors, warnings, and info messages is essential. If multiple modules create their own logger instances, it can result in duplicated log files or unorganized logging. Identify the key characteristics of the Singleton design pattern for a given pattern. (2)

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